Day 1: Is this an ML problem?

ML	Not ML

The first skill of a machine learning engineer is knowing when *not* to use machine learning. Today you will judge five real problems, decide whether ML belongs in each, and defend your choice.

Time	60 to 75 minutes
Work	Individual, written, no coding needed
Thinking level	Analyze and evaluate. You are judged on your reasoning, not just the verdict.
Submit	One page per problem, using the answer sheets at the end

Your tool: the seven-question checklist

Run every problem through these questions. In your justification, refer to them by number.

1	**Is there a known rule or formula?**	If the logic can be written down exactly, write it as code.
2	**Is there data that captures the pattern?**	Enough examples, with the right inputs recorded.
3	**Is an approximate answer acceptable?**	ML gives probabilities and estimates, never guarantees.
4	**Are labels or feedback available?**	This separates learning from known answers from finding structure on your own.
5	**Is the pattern stable over time?**	If the world shifts quickly, a model goes stale.
6	**What does a wrong prediction cost?**	Compare a false alarm with a missed case.
7	**Does a simple baseline already do the job?**	If an average or a rule is good enough, ML is hard to justify.

The five problems

For each one, decide **ML** or **Not ML**. If it is ML, name the task type. If it is not, say what you would use instead.

Problem 1 Payments

A payments app handles millions of UPI transactions a day. The fraud team wants to stop fraudulent transfers before the money moves.

What you have: Two years of transaction logs (amount, time, device, location, payee). Fraud is confirmed only when a customer complains, usually 2 to 3 weeks after the transaction. About 1 in 1,000 transactions is fraud.

Problem 2 Utilities

A state electricity board wants software that calculates each household's monthly bill from the units consumed. Bills must match the published tariff to the rupee, and every bill must be auditable.

What you have: The published tariff (slab rates, a fixed charge and duty) and the meter reading for each household.

Problem 3 Agriculture

A farmers' cooperative wants to advise members whether to sell their groundnut now or wait a week.

What you have: Five years of daily prices from the local mandi, daily arrival quantities, and rainfall records.

Problem 4 Dairy cooperative

A dairy cooperative has 8,000 member households. Management wants to offer different services (credit, cattle feed, training) to different kinds of members, but no member categories exist yet.

What you have: For every member, daily milk volume, fat percentage, payment history and herd size.

Problem 5 Exam cell

The exam section must prepare an invigilation duty roster: 120 faculty, 40 rooms, 6 sessions. No one may invigilate their own department's paper, no one may have a clash, and duties must be balanced to within one duty per person.

What you have: The faculty list with departments, the room list, the exam timetable, and last year's rosters.

Bonus problem Optional, for extra marks

The department wants to identify first-year students who are likely to drop out, so that mentors can reach them early.

What you have: Attendance, internal marks, fee status, hostel records, home region and gender for the last five batches.

Give a verdict, then answer: what action follows a prediction, who could be harmed by a wrong one, and what single fact would make you change your verdict?

No answer key is posted. A verdict that differs from your classmate's can still earn full marks if your reasoning is sound and uses the checklist. A verdict with weak reasoning will not.

Instructions for students

Read this page before you start. It explains how to attempt the challenge and how you will be marked.

Before you begin

- Finish the pre-reading guide and check that you can answer its six self-test questions.
- Keep the seven-question checklist in front of you while you work.
- Set aside one uninterrupted block of 60 to 75 minutes.
- You need a pen and the answer sheets at the end of this handout, or a document where you can copy the table.

How to attempt it

1	**Read all five problems once** without deciding anything. Notice what data exists and what is missing.
2	**Run each problem through the checklist.** Jot a quick yes or no against each of the seven questions.
3	**Decide your verdict:** ML or Not ML. If ML, choose the task type. If Not ML, say what you would use instead.
4	**Fill in one answer sheet per problem.** Keep every cell short and specific.
5	**Check your work.** Does each justification cite checklist numbers? Is your metric chosen for this problem, not copied from the last one?
6	**Try the bonus problem** only if you have time left.

Task types to choose from

Classification, regression, time-series forecasting, clustering, anomaly detection, rule or formula, optimization or constraint solving, or other (explain).

Rules

- Work on your own. You may discuss the checklist with friends, but write your own reasoning.
- You may be asked to explain any answer aloud.
- If you choose Not ML, write 'not applicable' for the predicted-target and metric rows, and use the space to explain your alternative.
- Typed and handwritten answers are both accepted. Handwriting must be readable.
- Late submissions are handled as announced by your instructor.

How you will be marked

Criterion	Marks per problem
Verdict is defensible and clearly stated	2
Task type is right, with reasoning	2
Metric and baseline suit the problem	2
Risks and failure modes are realistic	2
Checklist is used clearly in the justification	2
Total per problem	**10**

Five problems make 50 marks. The bonus problem adds up to 5 extra.

Submission

Submit by: _____

Submit to: _____

Name: _____ Roll number: _____ Batch: _____

Answer sheet: Problem 1

Name: _____ Roll no.: _____

Verdict (ML or Not ML)	
Justification (3 to 4 lines, cite checklist numbers)	
Task type, or what to use instead	
What is predicted, and from which inputs	
Success metric, and why accuracy is or is not the right one	
Simplest baseline to beat	
Biggest risk or failure mode	

Answer sheet: Problem 2

Name: _____ Roll no.: _____

Verdict (ML or Not ML)	
Justification (3 to 4 lines, cite checklist numbers)	
Task type, or what to use instead	
What is predicted, and from which inputs	
Success metric, and why accuracy is or is not the right one	
Simplest baseline to beat	
Biggest risk or failure mode	

Answer sheet: Problem 3

Name: _____ Roll no.: _____

Verdict (ML or Not ML)	
Justification (3 to 4 lines, cite checklist numbers)	
Task type, or what to use instead	
What is predicted, and from which inputs	
Success metric, and why accuracy is or is not the right one	
Simplest baseline to beat	
Biggest risk or failure mode	

Answer sheet: Problem 4

Name: _____ Roll no.: _____

Verdict (ML or Not ML)	
Justification (3 to 4 lines, cite checklist numbers)	
Task type, or what to use instead	
What is predicted, and from which inputs	
Success metric, and why accuracy is or is not the right one	
Simplest baseline to beat	
Biggest risk or failure mode	

Answer sheet: Problem 5

Name: _____ Roll no.: _____

Verdict (ML or Not ML)	
Justification (3 to 4 lines, cite checklist numbers)	
Task type, or what to use instead	
What is predicted, and from which inputs	
Success metric, and why accuracy is or is not the right one	
Simplest baseline to beat	
Biggest risk or failure mode	

Answer sheet: Bonus problem

Name: _____ Roll no.: _____

Verdict (ML or Not ML)	
Justification (3 to 4 lines, cite checklist numbers)	
Task type, or what to use instead	
What is predicted, and from which inputs	
Success metric, and why accuracy is or is not the right one	
Simplest baseline to beat	
Biggest risk or failure mode	